Explaining the Effects of Temperature, Wind Speed & Humidity

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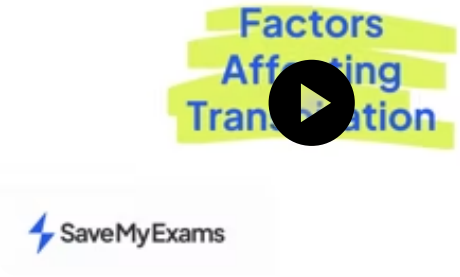

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Explaining the Effects of Temperature, Wind Speed & Humidity: Extended

Extended Tier Only

- Wind speed, humidity and temperature all have an effect on the rate at which transpiration occurs
- The table below explains how these factors affect the rate of transpiration when they are all high; the opposite effect would be observed if they were low

Transpiration Rate Factors Table

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Wind speed	High	More – good airflow removes water vapour from the air surrounding the leaf which sets up a concentration gradient between the leaf and the air, increasing water loss
Humidity	High	Less – humidity is a measure of moisture (water vapour) in the air; when the air is saturated with water vapour the concentration gradient is weaker so less water is lost
Temperature	High	More – at higher temperatures, particles have more kinetic energy so transpiration occurs as a faster rate as water molecules evaporate from mesophyll and diffuse away faster than at lower temperatures

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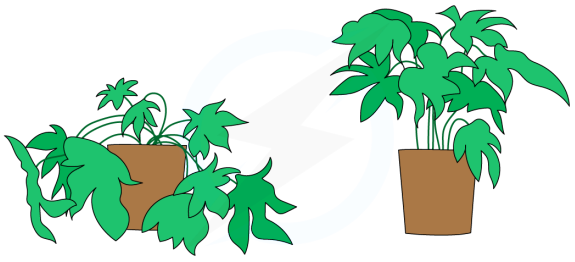
- A potometer can be used to investigate the effect of environmental factors on the rate of transpiration

Wilting: Extended

Extended Tier Only

- If more water evaporates from the leaves of a plant than is available in the soil to move into the root by osmosis, then **wilting** will occur
- This is when all the cells of the plant are not full of water, so the strength of the **cell walls cannot support the plant** and it starts to collapse



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A wilted plant cannot support itself and starts to collapse

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